

- 280.** All 2-digit numbers divisible by 5 are 10, 15, 20, 25, ..., 95.

This is an A.P. in which $a = 10$, $d = 5$ and $T_n = 95$.

$$T_n = 95 \Rightarrow a + (n - 1)d = 95 \Rightarrow 10 + (n - 1) \times 5 = 95$$

$$\Rightarrow (n - 1) = \frac{85}{5} = 17 \Rightarrow n = 18.$$

$$\therefore \text{Sum} = \frac{n}{2}(a + l) = \frac{18}{2}(10 + 95) = (9 \times 105) = 945.$$

- 281.** 3-digit numbers divisible by 6 are 102, 108, 114, ..., 996.

This is an A.P. in which $a = 102$, $d = 6$ and $T_n = 996$.

$$\therefore T_n = a + (n - 1)d \Rightarrow 102 + (n - 1) \times 6 = 996 \Rightarrow (n - 1) \times 6 = 894$$

$$\Rightarrow (n - 1) = 149 \Rightarrow n = 150.$$

Hence, there are 150 such numbers.

- 282.** Multiples of 7 between 11 and 200 are 14, 21, 28, 35, 42, ..., 189, 196.

$$T_m = 196 \Rightarrow 14 + (m - 1) \times 7 = 196 \Rightarrow (m - 1) \times 7 = 182$$

$$\Rightarrow (m - 1) = 26 \Rightarrow m = 27.$$

Multiples of 7 and 3 both, i.e. that of 21 are 21, 42, 63, ..., 189.

$$T_n = 189 \Rightarrow 21 + (n - 1) \times 21 = 189 \Rightarrow (n - 1) \times 21 = 168$$

$$\Rightarrow (n - 1) = 8 \Rightarrow n = 9.$$

$$\therefore \text{Required number of terms} = (27 - 9) = 18.$$

- 283.** Numbers between 14 and 95 and divisible by 3 are 15, 18, 21, 24, ..., 93.

$$T_m = 93 \Rightarrow 15 + (m - 1) \times 3 = 93 \Rightarrow (m - 1) \times 3 = 78$$

$$\Rightarrow (m - 1) = 26 \Rightarrow m = 27.$$

Numbers to be deleted are 33, 63, 93.

They are 3 in number.

$$\text{Required number of numbers} = (27 - 3) = 24.$$

- 284.** Required numbers are multiples of (10×13) , i.e. 130.

These numbers are 130, 260, 390, 520, 650, 780 and 910.

They are 7 in number.

- 285.** Number of integers between 100 and 150 (including both) = 51.

Numbers divisible by 3 are 102, 105, 108, ..., 150.

Let $T_m = 150$. Then, $a + (m - 1)d = T_m$.

$$\therefore 102 + (m - 1) \times 3 = 150 \Rightarrow (m - 1) \times 3 = 48 \Rightarrow m - 1 = 16 \Rightarrow m = 17.$$

Numbers divisible by 5 are 100, 105, 110, 115, ..., 150.

Let $T_n = 150$. Then, $a + (n - 1)d = T_n$.

$$\therefore 100 + (n - 1) \times 5 = 150 \Rightarrow (n - 1) \times 5 = 50 \Rightarrow (n - 1) = 10 \Rightarrow n = 11.$$

Numbers divisible by both 3 and 5 are 105, 120, 135, 150. They are four in number.

$$\text{Number of numbers divisible by 3 or 5} = (17 + 11 - 4) = 24.$$

$$\text{Number of numbers neither divisible by 3 nor by 5} = (51 - 24) = 27.$$

- 286.** Numbers from 501 to 599 which have 6 as digit are 506, 516, 526, 536, 546, 556, 560, 561, 562, 563, 564, 565, 566, 567, 568, 569, 576, 586 and 596, i.e. 6 occurs 20 times.

$$\text{Number of times 6 occurs from 600 to 699} = 100 + 20 = 120.$$

$$\therefore \text{Total number of times 6 occurs} = 20 + 120 = 140.$$

- 287.** Such numbers are 202, 212, 222, 232, 242, 252, 262, 272, 282, 292. There are 10 such numbers.

- 288.** Such numbers are 203, 213, 223, 233, 243, 253, 263, 273, 283, 293 and all numbers from 300 to 399. Clearly, number of such numbers = $10 + 100 = 110$.

- 289.** When the second digit is 1, third digit can be 0, i.e. there is one such number.

When the second digit is 2, third digit can be 0 or 1, i.e. there are 2 such numbers.

When the second digit is 3, third digit can be 0, 1 or 2 i.e. there are 3 such numbers, and so on.

When the first digit is 7, second digit can be 1, 2, 3, 4, 5 or 6. So, there are

$$(1 + 2 + 3 + 4 + 5 + 6) = 21 \text{ such numbers between 700 and 799.}$$

When the first digit is 8, second digit can be 1, 2, 3, 4, 5, 6 or 7. So, there are

$$(1 + 2 + 3 + 4 + 5 + 6 + 7) = 28 \text{ such numbers between 800 and 899.}$$

When the first digit is 9, second digit can be 1, 2, 3, 4, 5, 6, 7 or 8. So, there are

$$(1 + 2 + 3 + 4 + 5 + 6 + 7 + 8) = 36 \text{ such numbers between 900 and 999.}$$

$$\text{Hence, the required number} = (21 + 28 + 36) = 85.$$

- 290.** (a) In 183654729, 1836547 is not divisible by 7.

(b) In 381654729, 38 is divisible by 2, 381 is divisible by 3, 3816 is divisible by 4, 38165 is divisible by 5, 381654 is divisible by 6, 3816547 is divisible by 7, 38165472 is divisible by 8 and 381654729 is divisible by 9.

(c) In 983654721, 983 is not divisible by 3.

(d) In 981654723, 9816547 is not divisible by 7.

- 291.** On dividing 11109999 by 1111, we get:

So, the required remainder is 1110.

$$\begin{array}{r} 1111 \overline{) 11109999} (9999 \\ \underline{9999} \\ 11109 \\ \underline{9999} \\ 11109 \\ \underline{9999} \\ 11109 \\ \underline{9999} \\ 1110 \end{array}$$

- 292.** Number = $(68 \times 260) = 17680$. On dividing this number by 65, we get zero as remainder.

$$\begin{array}{r} 65 \overline{) 17680} (272 \\ \underline{130} \\ 468 \\ \underline{455} \\ 130 \\ \underline{130} \\ 0 \end{array}$$

- 293.** $(2176 - 9) = 2167 = (11 \times 197)$.

So, the required number is 197.

294. (a) $\begin{array}{r} -7 \overline{) -112} (16 \\ \underline{-112} \\ 0 \end{array}$

(b) $\begin{array}{r} -9 \overline{) 118} (-13 \\ \underline{-9} \\ 28 \\ \underline{-27} \\ 1 \end{array}$

(c) $\begin{array}{r} 6 \overline{) -109} (-19 \\ \underline{-6} \\ -49 \\ \underline{-54} \\ 5 \end{array}$

(d) $\begin{array}{r} 8 \overline{) 115} (14 \\ \underline{8} \\ 35 \\ \underline{32} \\ 3 \end{array}$